

JOSEPH CUFFNEY

josephcuffney@gmail.com · 585-503-2714 · Rochester, NY
github.com/jcuffney · linkedin.com/in/cuffney

Professional Summary

Senior software engineer with 10+ years shipping fast from problem to production — launching solo in 5 weeks at Nike Valiant Labs, and leading a 10-engineer team to take Coinbase's capital formation platform live in 2 months. Now building LLM agent systems end to end: Claude and Bedrock, MCP and tool use, RAG, NLP routing, and evals.

Experience

Coinbase · Senior Software Engineer

January 2022 – Present · Remote

Capital Formation

- Led a 10-engineer team to launch a new capital formation platform supporting IPOs alongside token sales.
- Set technical direction and delivered end to end — built the whole product and went to production in 2 months.
- Raised over \$250M for the platform's first issuer.

GraphQL Federation

- Led a company-wide data-layer re-architecture and migration — designed a federated GraphQL architecture to replace the monolith and set direction for teams migrating onto it, improving build times and reliability.
- Led development of the first subgraph (Golang, gqlgen), mentoring engineers across teams as they built their own.
- Deployed federated GraphQL services on Kubernetes for scaling and cost optimization.

INTX Exchange

- Planned, built, and launched perpetual futures — a net-new retail asset type — across product and platform teams.
- Owned monitoring, scaling, reliability, and incident response during the global rollout.
- Drove over \$1B in trading volume in the product's first year live.

Nike Valiant Labs · Sr. Software Engineer III

June 2018 – December 2021 · Portland, OR / Remote

Cultivator

- Built an early iteration of what became Nike By You — a creator-driven sneaker marketplace where 100+ creators sold 500k+ pairs for charities of their choice, converting at 13% against a ~1% industry standard.
- Designed the marketplace's microservice architecture and API: API Gateway fronting AWS Lambda and ECS.
- Implemented a CI/CD pipeline with AWS CodePipeline and CodeBuild to speed up delivery.

Nike Valiant Labs · Lead Software Engineer (contract)

August 2017 – June 2018 · Portland, OR

Tandem Personal Training

- Sole engineer on a Nike Valiant Labs venture — took it from zero to launch in 5 weeks, owning architecture, build, payments, and infrastructure, and working directly with business, marketing, and design partners.
- Designed and built a subscription service connecting clients with personal trainers in the Portland metro — Craft CMS on Elastic Beanstalk, with AWS Lambda handling payments and Stripe webhooks.
- Implemented a two-party Stripe payment system: subscription invoicing, transfers, refunds, and renewals.
- Ran content iterations and A/B tests with marketing and design to increase conversion rates.

The Program · Technical Lead

October 2015 – August 2017 · Portland, OR

- Owned client projects end to end across React/Redux, Express, Spring Boot, and Craft — scoping, estimation, and sprint planning through delivery, while contributing as a primary developer on others.

- Partnered with clients and project managers on process, and mentored engineers on React/Redux.
- Owned developer operations — environments and CI — using Jenkins, Ansible, Docker, and DigitalOcean.

Projects

Jarvis Cloud · Voice agent on AWS · 2026

- Voice assistant on Amazon Nova 2 Sonic that delegates deeper work to Claude agents on Bedrock via tool use.
- Wrote an MCP client (stdio and Streamable HTTP) and integrated Home Assistant over WireGuard, Gmail, Google Calendar, and S3 Vectors document knowledge, with a monthly spend cap on model costs.
- Built in two weeks (338 commits): TypeScript and React on CloudFormation, deploying on every push, plus Claude Code jobs dispatched to a LAN devbox through a long-poll work queue.

Assistant · Local-first agent in Rust · 2026

- Rule-based NLP triage, then a ModernBERT classifier labels route, tone, and task — code, not the model, decides.
- Planner → researcher → reporter agents; policies proceed autonomously but escalate to a human when required.
- Built in two days — 12k+ lines of Rust across 9 crates — with an MCP client (rmcp), SearXNG search, guarded web fetch, labeled routing evals, and evals on the CLINC150 intent benchmark.

Orca(stration) AI · From The Forest · 2026

- Building a platform that makes AI agents first-class members of the org chart — Rust microservices on Kubernetes.
- Designed an agentic reasoning loop — autonomous within guardrails, escalating to a human beyond them.
- Built AI budgeting on x402 micropayments, keeping agent spend within budget and ROI measurable.
- Extended LDAP so humans and agents share one directory — teams adopt agents without changing how they work.

LLM Gateway & Homelab Platform · 2026

- LiteLLM gateway: local llama.cpp first, Claude on Bedrock as fallback, per-consumer keys and spend tracking.
- Talos Kubernetes on Proxmox via Terraform and Argo CD, a GPU shared in 4 GB slices, CI-only tenant deploys.
- Self-hosted speech-to-text and text-to-speech, chosen by measured time-to-first-byte and realtime factor.

On Set! · Set-location discovery app · 2020 – 2022

- Co-founded a film set-location app: 2,000+ locations mapped, 200 small-business partners, 1,000 MAU in year one.
- Built the web and React Native apps on an Apollo Federation GraphQL API over AWS Lambda and DynamoDB.

Skills

AI & AGENTS

Claude / OpenAI / Grok Claude Agent SDK MCP Agent orchestration RAG Evals

HARNESSES

Claude Code Codex OpenCode Hermes

NLP

Intent classification (ModernBERT) Semantic analysis & search Embeddings Speech (STT/TTS)

MODELS

AWS Bedrock LiteLLM llama.cpp Ollama Tool use Prompt & tool design

STACK

TypeScript Python Go Rust React React Native Node.js

APIs

GraphQL (Federation) gRPC REST Stripe Webhooks

CLOUD & INFRA

AWS Kubernetes Argo CD Terraform CloudFormation Proxmox Docker

DATA & TOOLS

Postgres SurrealDB DynamoDB Elasticsearch n8n Datadog Snowflake

Education

Rochester Institute of Technology · Rochester, NY

Bachelor of Science, Computer Science